



**Technical Specifications
for
2x20 MW Solar PV Plant Project
in
AGOGA and MISOZI
UGANDA**

1. SOLAR POWER PLANT

This general scope of work involves Engineering, Procurement and Construction (EPC) of the grid-connected solar photovoltaic power plant with a capacity of **20 MW**.

The scope of work includes commissioning of the power plant. The Contractor shall construct and commission plant to deliver guaranteed plant performance.

Work includes Comprehensive Operation and Maintenance (O&M) of the plant.

2. CIVIL WORK:

Design, fabricate, supply and installation of module mounting structures; civil footings / foundations of appropriate and anchoring of fixed tilt module mounting structure; work required for site development activities inclusive but not limited to area grading, excavation, cutting, filling and disposal of soil from site, leveling of land; RCC cable trenches & storm water drainage system, construction of room for inverter with electrical fixtures and proper air circulation, provide pipeline network with necessary control panel, pipe line, taps, flexible hoses etc. for washing of modules and gardening

3. MODULE MOUNTING STRUCTURE

3.1 The technical specifications for mounting structures are as under :-

- a. Support structure design and foundation or fixation mounting arrangements should withstand horizontal wind speed as per relevant IS Code.
- b. The total load for roof top SPV should be less than 60 Kg/m².
- c. The load bearing capacity and residual life of the support structures must be checked and measures to strengthen structure, if required and economical, be taken.
- d. The module mounting steel structure shall be **Mild Steel Fixed Tilt** as per IS 2061- 1992 and galvanizing **HDGI (Hot dip Galvanized) with 85 micron coating** as per IS 4759 and should be designed as per the requirement of the installations and module configurations.
- e. All fasteners, nut and bolts should be made of Stainless steel - SS 304 Grade, painted with rust free paint and should allow easy replacement of any module.
- f. The minimum clearance of the structure from the roof level should be 300-400 mm.
- g. The material shall be corrosion resistant and electronically compatible with material used in module frame.
- h. The structures shall be designed to allow easy replacement of any module. Each structure should have angle of inclination as per the site conditions to take maximum insulation.
- i. Modules shall be mounted towards south facing and at a suitable inclination (28°) to maximize annual energy output.

3.2 FOUNDATION BASE

The legs of the structures should be made with GI Channels and fixed with foundation as described in provisional lump sum.

- a. Design calculation shall be done through STADD analysis and contractor shall submit the design review and approval of engineering in-charge.

4. DC DISTRIBUTION BOARD (DCDB)

- a. DCDB shall be provided in between PCU and Solar Array. It shall have MCCB of Suitable rating for connection and disconnection of array section. It shall have meters for measuring Array voltage and Array current.
- b. Type II Surge Arrestor should be incorporated for surge protection as per IEC 60364-5-53 with meters for measuring the array voltage and array current.

- c. The DC distribution box shall be of IP65 rating made of thermo plastic/CRCA sheet.
- d. The junction boxes shall have suitable cable entry points with cable glands of appropriate sizes for both incoming and outgoing cables.
- e. Suitable markings should be provided on the bus bar for easy identification, and cable ferrules shall be fitted at the cable termination points for identification

5. SOLAR CELLS AND PV MODULES

The technical specifications of PV Module for use in Grid Solar Power plants will be:-

- a. Supply, installation, testing & commissioning of PV modules shall be on the Vendors scope.
- b. The PV Modules used in the solar power projects must qualify to the latest edition of the International Electrotechnical Commission (IEC) PV module qualification test or equivalent BIS standards.
- c. Solar PV minimum array capacity should not be less than **20 MW** and should comprise of solar crystalline modules at standard test conditions, indigenous manufactured Solar PV Crystalline Silicon Terrestrial PV Modules IEC 61215.
- d. The modules must conform to IEC 61730.
- e. PV modules to be used in a highly corrosive atmosphere (coastal areas, etc.) must qualify Salt Mist Corrosion Testing as per IEC 61701.
- f. PV modules to be used in a sandy environment must qualify the IEC 60068-2-68 standards.
- g. PV modules used in grid connected solar power plants must be warranted for output wattage, which should not be less than 90 % at the end of 10 years and 80 % at the end of 25 years.
- h. The mechanical structures, electrical works and overall workmanship of the grid solar power plants must be warranted for a minimum of 5 years.
- i. The V-I curve of each PV module with Sl. Nos. should be submitted along with modules meeting the required specifications.

6. TEST CENTRES

- a. The PV modules must be tested and approved by one of the IEC authorized test centers. In addition, a PV module qualification test certificate as per IEC standard.

7. IDENTIFICATION AND TRACEABILITY

Each PV module must use a RF identification tag (RFID). The following information must be mentioned in the RFID used on each module. This must be laminated inside the panel and it must be able to withstand harsh environmental conditions.

- (i) Name of the manufacturer of PV Module
- (ii) Name of the Manufacturer of Solar cells
- (iii) Month and year of the manufacture for solar cells and module individually
- (iv) Country of origin for solar cells and module individually
- (v) I-V curve for the module
- (vi) Peak Wattage, Im, Vm and FF for the module
- (vii) Unique Serial No and Model No of the module
- (viii) Date and year of obtaining IEC PV module qualification certificate
- (ix) Name of the test lab issuing IEC certificate
- (x) Other relevant information on traceability of solar cells and module as per ISO 9000 series.

8. JUNCTION BOXES

- a) General Requirements IP 65 as per IEC 529
- b) The junction boxes are to be provided in the PV array for termination of connecting cables. The J. Boxes (JBs) shall be made of GRP/FRP/Powder Coated Aluminium /cast aluminium alloy with full dust, water & vermin proof arrangement.
- c) All wires/cables must be terminated through copper cable lugs. The JB's shall be such that input & output termination can be made through suitable cable glands.
- d) Copper bus bars/terminal blocks housed in the junction box with suitable termination threads Conforming to IP65 standard and IEC 62208 Hinged door with EPDM rubber gasket to prevent water entry.
- e) Single compression cable glands. Provision of earthings. It should be placed at 5 feet height or above for ease of accessibility.
- f) Each Junction Box shall have High quality Suitable capacity Metal Oxide Varistors (MOVs) / surge arrestors, suitable Reverse Blocking Diodes.
- g) The Junction Boxes shall have suitable arrangement monitoring and disconnection for each of the groups.
- h) Suitable markings shall be provided on the bus bar for easy identification and the cable ferrules must be fitted at the cable termination points for identification

9. TRANSFORMER

9.1 Dry/oil type along with all protections, switch gears, Vacuum circuit breakers, cables etc. along with required civil work The transformer shall conform in all respects to the relevant Indian/International standards specifications with latest amendment thereof, some of them are listed below:-

Title	International Std.
Specification for Power transformer	IEC – 76
Insulating Oil for transformer	BS –148
Fittings and accessories for power	ASTM D-1275
High voltage porcelain bushings	IEC 296-1969
Low voltage porcelain bushings	
Dimensions for outdoor bushings	DIN 42531 to 33
Specification for copper wire rods	ASTM B - 49
Specifications for colors for ready mixed paints	
Guide for loading oil immersed	
Climate Proofing	
Degree of Protection	

9.2 Transformers should have all necessary protection features like silica gel breather, Buchholz relay, pressure relief device, magnetic oil level gauge, oil temperature indicator and winding temperature indicator

- Checking Oil and winding temperature and colors of silica gel in the breather & Cleaning monthly
- Testing of main tank oil for BDV and moisture content on quarterly basis.
- Lubricating/Greasing all moving parts of OLTC mechanism. Measurement of IR values with suitable megger according with rating.
- Physical examination of diaphragm of vent pipe for any crack.
- Testing OLTC oil for BDV and moisture
- Testing of Oil for dissolved gas analysis, acidity, tan-delta, interface tension.

- Measurement of DC winding resistance
- Turns ratio test at all taps
- Over haul of tap change mechanism
- Changing the gasket set allocation as when leakage is found or gasket damaged

9.3 TYPE & TEST CERTIFICATES

(A) TYPE TESTS: -

- (i) All the equipment offered shall be fully type tested as per relevant technical specification.
- (ii) The following type tests shall be carried out on the transformers in accordance with relevant technical standards: -
 - (a) Short Circuit Test.
 - (b) Lightening Impulse Voltage withstand test.

The rating and terminal marking plate shall exhibit the following details:

Temperature rise test.

1. Make of Transformer.
 2. Manufactured by M/s.
 3. Type of Transformer - Conventional Type
 4. Specification Ref. No.
 5. KVA Capacity (Conv.)
 6. Type of cooling ONAN.....
- Volts at HV Frequency Hz 50
 No Load LV Impedance Volts %
 No Load Amperes HV Vector group
 Ref. LV Core & winding wt. in Kgs.
 Phases HV Wt. of oil in Kgs.
 LV Total wt. of X-mer in Kgs.
 Makers S. No.
 Oil in litres.
 Date of dispatch
 Untanking Wt. in Kgs.
 Guarantee maximum temperature rise in oil 45°C.

9.4.1. **INDOOR TYPE INVERTER:**

One number of CENTRAL INVERTER UNIT

The output of the inverter system is to be connected with the grid system. The contractor should select the inverter (Central) as per its own system design so as to optimize the power output.

The technical specifications for inverter are as :-

- | | | |
|--------|----------------------------|---|
| (i) | Nominal Voltage | 3 phase, 50 Hz |
| (ii) | Control | Microprocessor/ DSP |
| (iii) | Switching Device | IGBT / MOSFET |
| (iv) | Grid freq. Synchronization | +/- 3 Hz or more |
| (v) | Ambient temperature | -20°C to +50° C |
| (vi) | Humidity | 95 % non- condensing |
| (vii) | Protection of enclosure | IP 20 (Indoor rated)/ IP 65 (Outdoor rated) |
| (viii) | No load losses | < 1% of rated power |
| (ix) | Inverter Efficiency | >97 % (without inbuilt galvanic isolation)
>93 % (with inbuilt galvanic isolation) |
| (x) | Power Factor Control Range | >0.9 lead or lag |
| (xi) | THD | < 3 % (at rated power) |

- (xii) Maximum Input voltage THD1000 V DC
- (xiii) Maximum Noise Level 75 dB DC injection Less than 0.5% of nominal load current 75 dB standards.
- (xiv) Inverter should have internal protection arrangements against any sustainable fault in feeder line and against the lightening on feeder.
- (xv) Should have insulating transformer.

9.4.1.2. Inverter/ Power Condition unit must provide protection against:

- Over current
- Sync loss
- Over temperature
- DC bus over voltage
- Cooling Fan failure (If provided)
- Short circuit
- Lightning
- Earth fault
- Surge voltage induced at output due to external source
- Power regulation in the event of thermal overloading
- Set point pre-selection for VAR control
- Bus communication via -interface for integration
- Remote control via telephone modem or mini web server
- Integrated protection in the DC and three phase system
- Insulation monitoring of the PV array with sequential fault location

Ground fault detector which is essential for large PV generators in view of appreciable discharge current with respect to ground.

Over voltage protection against atmospheric lightning discharge to the PV array is required.

The power conditioner must be entirely self-managing and stable in operation.

A self-diagnostic system check should occur on start up. Functions should include a test of key parameters on start up.

9.4.1.3. Earthing of Inverters:

DC side of each inverter shall be earthed to distinct earth pit through adequate size conductor. The size of conductor shall be as per the maximum fault current of DC system.

9.4.2. **Technical requirements of PCU**

PCU shall confirm to the following standards:

- | | | |
|-------|-------------------------------|--|
| i. | Efficiency measurement: | IEC 61683 |
| ii. | Environmental Testing: | IEC 60068-2 |
| iii. | Product safety: | IEC 62109 (1&2) |
| iv. | Grid connectivity Interface: | IEC 61727: 2004 |
| v. | Interconnection PV Invertors: | IEC 62116 |
| vi. | Electromagnetic Interface : | IEC 610002, 3, 6 |
| vii. | Ingress Protection : | IP 54 & IP 21 |
| viii. | Batteries: | As per relevant BIS Std. |
| ix. | Cables : | IEC 60227 |
| x. | Earthing / Lightening : | IEC 62561 |
| xi. | Surge Arrestors : | IEC 60364-5-53 |
| xii. | Circuit Breakers : | IEC 60947 1, 11, III |
| xiii. | Junction Boxes & Enclosures : | IP 65/ IP 21 |
| xiv. | Meters : | As per DISCOM guidelines |
| xv. | Grid Connectivity : | As prevalent in the grid code |
| xvi. | EMC, harmonics, etc.: | IEC 61000 series, 6-2, 6-4 and other relevant Standards. |

- xvii. Recommended practice for PV – Utility interconnections:
IEEE standard 929 – 2000 or equivalent
- xviii. Protection against islanding of grid: IEEE1547 or equivalent
- xix. Reliability test standard: IEC 62093 or equivalent

9.4.2.1. Desired Technical Specifications of PCU

- Sinusoidal current modulation with excellent dynamic response.
- Compact and weather proof housing (indoor/ outdoor)
- Comprehensive network management functions (including the LVRT and capability to inject reactive power to the grid)
- Total Harmonic Distortion (THD) < 1% of rated power and maximum loss in sleep mode shall be less than 0.05%
- Optional VAR control
- Unit wise & integrated Data logging
- Dedicated Prefabs / Ethernet for networking

9.4.2.2. PCU/inverter front panel shall be provided with display (LCD or equivalent) to monitor, but not limited to, the following:-

- i. DC power input
- ii. DC input voltage
- iii. DC Current
- iv. AC power output
- v. AC voltage (all the 3 phases and line)
- vi. AC current (all the 3 phases and line)
- vii. Power Factor
- viii. Inverter on
- ix. Grid on
- x. Inverter under voltage/ over voltage
- xi. Inverter over load
- xii. Inverter over temperature

9.4.2.3 Documentary Requirements & Inspection

- i. The bill of materials associated with PCU's should be clearly indicated while delivering the equipment.
- ii. The Contractor shall provide to the Employer, data sheet containing detailed technical specifications of all the inverters and PCUs, Type test reports and Operation & Maintenance manual before dispatch of PCUs.
- iii. The Employer or its authorized representative reserves the right to inspect the PCUs/ Inverters at the manufacturer's site prior to dispatch

9.5. METERING – Energy accounting

- a. Contractor shall provide ABT compatible interface meters as main meters based on voltage, point and period of supply and tariff category.
 - i. As per provisions of Grid code., two sets of high precision ABT compatible energy Meter/Import-Export Meter with associated set of metering equipment are to be installed at generating point
 - ii. For 33Kv generator, 2 Nos. Of combined CT-PT units of Class 0.2s accuracy shall be provided by the developer at point of injection to serve to main meters and check meters for billing purpose, at developers cost.
 - iii. The developer has to get the equipment tested before installation at site.
 - iv. The GM shall have to certify the meters, metering equipment as per grid standards in consultation with meter testing division before installation at site.

- v. Main meter shall always be maintained in good condition at developers cost.
 - vi. Main meters shall always be maintained in good condition and shall be open for inspection by authorized entities.
 - vii. The concerned licensee may provide check meters of the same specifications as main meters.
 - viii. The main and check meters shall be periodically tested and calibrated by the concerned licensee in the presence of other party involved. Main and check meters shall be sealed by both parties. Defective meter shall be replaced immediately.
 - ix. Reading of main and check meters shall be taken periodically at appointed day and time by authorized officer of the concerned licensee, the generator and the customer or his representative, as the case may be.
 - x. Readings of the check meters shall be considered for billing purposes when main meters are found to be defective or stopped.
 - xi. Both the main meter and check meter shall be tested for accuracy if difference between the readings of main and check meters vis-à-vis main meter reading exceed twice the percentage errors permissible for relevant accuracy class.
 - xii. If an open access customer requires the licensee to provide main meters he shall provide security deposit to the licensee and shall pay for hiring charges. The meter shall be maintained by the licensee.
 - xiii. The location of meter and height of meter display from floor shall be as per Standard Code of Practice.
 - xiv. For outdoor installations, the meters shall be protected by appropriate enclosure of level of protection.
 - xv. Limits of error as per International standards.
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- b. Capability of recording frequency in every 15 minutes. Load survey data of meter may be verified.
 - c. Capability of meter for recording of Reactive High & Reactive low.
 - d. Recording of Import and Export energy for every 15 Min.
 - e. Recording of mid night energy for Active and Reactive Parameters.
 - f. For Metering Equipment:
 - i) 33 KV CT
 - a. Accuracy test (determination of Error)
 - b. Verification of terminal polarity.
 - c. HV test on secondary winding.
 - ii) 3 phase 33 KV PT
 - a. Accuracy test (determination of Error).
 - b. Verification of terminal /polarity.
 - c. High Voltage power frequency test on secondary winding.
 - d. Induced over voltage test.

9.6. SCADA – Data Monitoring & Acquisition

- (i) SCADA system of PCU shall clearly depict the single line diagram of the plant on the monitor. Mimics shall be provided with radio buttons to show the following parameters:
 - a. Solar radiation (W/m²)
 - b. PV module back surface temp

- c. Ambient temp
- d. Inverter output power (3 phase)
- e. Line and phase currents
- f. Cumulative energy exported

- (ii) SCADA system shall perform following control operations :-
 - a. Inverter ON/OFF
 - b. Set point editing through a proven password mechanism
 - c. Mimic control through PC keyboard operation

- (iii) SCADA shall also conform to following features :-

SCADA system shall incorporate integrated system control and data acquisition facilities.

The use of a local operator interface and latest technology features shall be incorporated to enable viewing of instantaneous parameter metering, changing of operator modes and review of system logged events.

Further, with PC based latest software technology, solar plant shall be monitored remotely via satellite link.

The major SCADA features incorporated in to the control system are listed below.

Operator interface of latest technology: Instantaneous grid, array, inverter, AC, and metering of all parameters.

- a. Integrated AC, DC data point logging: Instantaneous logging of all parameters. Including AC parameters, generator run hours and energy details.
- b. Fault and system diagnostics with time stamped event logging.
- c. Selectable event logging resolution for enhanced diagnostics.
- d. Remote SCADA features with specific needs of station monitoring and remote communication.
- e. Remote system access software, secured transmission of data and central PC facility.

9.7. LT PANEL

- i. LTP DB shall have EDO ACBS (Electrically operated Draw-Out Air Circuit Breaker) for ensuring necessary isolation between PCU and transformer.
- ii. ACB is microprocessor based electronic trip (ET) systems with thermal memory, over load, short circuit protection and also has protection against earth leakage faults.
- iii. LTPDB also equipped with Transient Voltage Surge Suppressor (TVSS) for protection against transient voltage surges.
- iv. Digital Ammeter and Digital Voltmeter with their inbuilt selector switches and Digital Load manager which performs as a multi-function meter shall be provided on the front side of the cubicle.
- v. LTPDB is also provided with a Breaker Control Switch for electrically operating the ACB. NO and NC contacts for ACB are provided to communicate the On, OFF and Trip status to SCADA.

9.8. HT PANEL

- i. The 33 KV from HV side of the transformer shall be fed to grid with proper metering, protection and switch-gear which can be done by Metering, Relays and Circuit Breakers located in HT panels. In power plants at all three locations, one outgoing VCB to grid and one Bus PT panel for measurement of bus voltage and protection against abnormal bus voltages.
- ii. All these HT panels shall be combined with a common bus bar which is internally connected among these panels.

- iii. These HT panels shall be mounted on a well-constructed and raised platform located in switch yard area. In addition to these HT panels, a Remote Annunciation Panel (RAP) shall be provided for remote monitoring and control of HT panels which is located inside control room.
- iv. Each Vacuum Circuit Breaker (VCB), Current transformer (CT), Relays and Metering equipment like Digital Ammeter and Load Manager. Bus PT panel comprises of Potential Transformer, Over voltage and under voltage relay and Digital Voltmeter.
- v. Enough space shall be provided on either side of the HT panel for future expansion.
- vi. **Protective Relays:-**
 - a) Over current and Earth fault relay: A microprocessor based numerical relay for protection against instantaneous/ IDMT faults like over current, short circuit current, over load, earth faults and instantaneous earth faults should be used.
 - b) Master Trip Relay: This relay is the main relay which activates the trip circuit of the breaker upon the trip signals received by other relays.
 - c) Trip Circuit Supervision Relay: This relay is also of electro mechanical type and this supervises whether the trip circuit is healthy or not.
 - d) DC Failure Relay: Whenever DC supply fails to HT panels, this relay gets activated and indication will be given in HT panels and alarm and indication will be given in RAP.
 - e) Transformer Alarm Relay: This auxiliary relay is meant for Transformer protection and gives alarm against faults occurred in transformer such as Buchholz relay alarm, OTI alarm and WTI alarm.
 - f) Transformer Trip Relay: This relay is for actuating the breaker to trip against trip signals from transformer for Buchholz relay trip, WTI trip and OTI trip.
 - g) Transformer Alarm/Trip Relay (Tat): This relay is meant for both alarm and trip signals from transformer against PRD alarm, PRD trip and LOLA alarm faults.
 - h) Metering Equipment: Digital Ammeter with inbuilt selector switch and Load Manager shall be provided in incomers and outgoing panels.
 - i) Bus PT Panel: Bus PT panel is a separate panel adjacent to incomers and outgoing feeders, connected to bus for measuring the bus voltage and to protect against abnormal bus voltages. The core meant for measurement is connected to Digital Voltmeter with inbuilt selector switch and other core which is protection core is connected to Over voltage/Under voltage relay.
 - j) **Remote Annunciation Panel (RAP):**
 - RAP is provided to monitor and control the HT panels remotely i.e., from control room to ensure the reliability and safety for the operating personnel. This cubicle shall have number of windows for displaying all types of faults and those windows turns red on fault. Out of these windows, there will be separate indications meant for Incomer -1, Incomer -2, for outgoing panel, for Bus PT and some left as spare.
 - RAP also equipped with buzzers to give alarm on faults. Out of these, one works on DC supply and other works on AC supply.
 - RAP shall have Indication lamps each for incomers and outgoing panels to indicate its status like VCB On, VCB Off, Test position, Service position Trip circuit healthy, DC fail and Spring charged condition. RAP will also have three Breaker Control Switches (BCS) for remote control of incomers and outgoing HT panels.
- vii. VCBs of HT panels can be tripped or closed using these switches. But, the operation of these T-N-C switches depends on the electrical interlocks that shall be made among the incomers and outgoing panel

viii. **Earthing Truck:**

Earthing trucks are provided along with the HT panels, even though these are not a part of power flow in the plant, because these panels are meant for protecting the working personnel performing maintenance on HT panels. When operator performs maintenance works on bus bar of HT panels gets electrified if any one closes the downstream breaker. So, to avoid such instances, earthing trucks for bus bar earthing are provided in the place of outgoing breaker to earth the power supply immediately and it also produces alarm and indication for operator.

9.9. EARTHING SYSTEM

- a. Each array structure of the PV should be grounded properly.
- b. All metal casing/shielding of the plant should be thoroughly grounded.
- c. All the earthing in the plant shall comply with relevant international standards.
- d. A minimum of two separate dedicated and interconnected earth electrodes must be used for the earthing of the solar PV system support structure.
- e. The earth electrodes shall have a pre-cast concrete enclosure with a removal lid for inspection and maintenance. The entire earthing system shall comprise non-corrosive components.

9.10. LIGHTENING ARRESTORS

- a. The lightening arrestor shall be heavy duty station class type, discharge class III, conforming to IEC specification
- b. The required number of suitable lightning arrestors shall be installed in the array field.
- c. Lightning protection shall be provided by the use of metal oxide varistors and suitable earthing such that induced transients find an alternate route to earth.
- d. Concrete foundation for holding the lightning conductor, in position, to be made after giving due consideration to maximum wind speed and maintenance requirement at site in future.

9.10.1. Cables and Wires

All cables and connectors to be used for installation of solar field must be of solar grade which can withstand harsh environment conditions for 25 years and voltages as per latest IEC standards.

9.10.2. DC CABLE

The DC Cables in a solar PV plant to be used in the following areas:-

- a. Interconnecting SPV modules
- b. From SPV Modules up to String Monitoring Unit(SMU)
- c. From SMU up to the Inverter

9.10.3. HT CABLE

HT cable should meet the international standard specifications and requirements.

LT CABLE

a. Cabling on AC side (LT)

LT Power Cable shall be armoured, unearthed grade, double core/multi-core, stranded Cu/Al conductor as mentioned, XLPE insulated with PVC outer sheath made on PVC compound.

b. All electrical cables / wires to be laid/fixed in Conduit.

c. Proper laying of cables shall be ensured in appropriate cable trays, pipes/ trenches as per site requirement.

9.11. SURGE PROTECTION DEVICES

Surge Protection Devices for protection against surge currents and voltages shall be provided. Internal surge protection shall consist of three MOV type surge-arrestors connected from +ve(positive) and –ve(negative) terminals to earth.

9.12. CONNECTORS

Connectors should have the specifications as under:-

1000V (IEC/CEI),600V (UL), with Minimal voltage drop, High current-carrying capacity, Minimal power loss, Minimal contact resistance, High durability contacts withstand up to 1 million mating cycles, Operating temperatures up to 350°C,higher temperatures permitted for short periods.

9.13. CCTV

CCTV Cameras along with monitoring stations (sufficient numbers) and all other accessories required for its proper operation must be installed to have complete coverage of the following areas :-

- (i) Main entry: Covering all the entry/exit for 24 hrs
- (ii) Along the Plant Perimeter: Covering complete perimeter of Plant Area to capture all possible intrusion
- (iii) Control Rooms: Covering Entry/Exit and activities within Control Rooms. Contractor has to propose the locations and number of cameras required for the Plant during bidding, however Employer's decision on number of cameras shall be final.
- (iv) Monitoring stations of the CCTV Network shall be installed in Main Control Room.
- (v) The CCTV system shall be designed as a standalone IP based network architecture. System shall use video signals from different cameras at defined locations, process the video signals for viewing on monitors at control room and simultaneously record all video streams using latest compression techniques.
- (vi) Camera shall be colour, suitable for day and night surveillance (even under complete darkness) and network compatible.
- (vii) It should be possible to control all cameras i.e., PTZ auto/ manual focus, selection of pre-sets, video tour selection etc. The software shall support flexible 1/2/4 windows split screen display mode or scroll mode on the display monitor for live video.
- (viii) The system shall support video analytics in respect of the following:
 - a. Video motion detection

- b. Object tracking
- c. Object classification
- (ix) Camera server shall be provided with sufficient storage space to storage recordings of all cameras at HD mode for a period of 15 days. All recordings shall have camera ID, location, date and time of recording

9.14. WEATHER MONITORING STATION

The contractor is required to install necessary equipment to continuously measure the Solar Radiation and other climatic conditions through a weather monitoring station.

The major categories of site-specific assessment required are:

- a. Global Solar Radiation ("GSR")
- b. Diffuse Solar Radiation ("DSR")
- c. Sunshine Duration
- d. Atmospheric Turbidity
- e. Temperature & Humidity
- f. Wind Speed

Simultaneously, measurement of the generation of DC power as well as AC power from the plant must be done. Such measurement facility should be placed as close as possible to the solar plants. A common measurement facility may be allowed to be used by the plants located within a radius of five km from this facility.

9.16 DANGER PLATES

Size of each Danger Notice plates shall be 200 mm x 150 mm made of mild steel sheet and at least 2 mm thick, and vitreous enamelled white on both sides and with inscription in signal red colours on front side as required.

9.17. POWER EVACUATION

The evacuation voltage shall be at 33 kV AC (three phase) wherein evacuating point cum metering point shall be installed at 33 kV interconnection point within the boundary of solar plant.

ABT meter to measure net power evacuation shall be installed at 33 kV interconnection point on four pole structure near solar PV plant boundary.

9.18. OPERATION & MAINTENANCE

Contractor shall be responsible for comprehensive operation & maintenance of the plant. After successful operation of plant for **11 MONTHS** after commissioning towards performance demonstration, as detailed in technical specification including supply and storage of all spare parts, consumables, repairs/ replacement of any defective equipment etc.

- i. Deputation of qualified and experienced engineers for taking care of O&M activities.
- ii. Cleaning of Solar Module on weekly basis and wipers can be used to remove the accumulated dirt and bird droppings.
- iii. Task of the operators is to check periodic logging of the voltage, current, power factor, power and energy output of the plant. The operator shall monthly record the energy output, down time, etc.
- iv. Monitoring the performance of the solar power Plant and regular maintenance of the whole system including Modules, PCU's, transformers, overhead line, outdoor/indoor panels/ kiosks etc. are necessary for extracting and maintaining the maximum energy output from the solar power plant.

9.19. WARRANTY

The mechanical structures, electrical works including power conditioners/ inverters/ charge controllers/ maximum power point tracker units/ distribution boards/ digital meters/ switch gear/ storage batteries, etc. and over all workmanship of the PV power plants/ systems must be warranted against any manufacturing/ design/ installation defects for a minimum period of 5 years.



MISSOZI SITE UTM COORDINATES		
	EASTINGS	NORTHINGS
1	343526.825	41093.336
2	343497.348	39888.031
3	344578.015	39856.817
4	344602.755	40691.469
5	344451.105	40939.901
6	344451.105	40939.901
7	344451.105	40939.901
8	343919.478	41214.468
9	343919.478	41214.468
10	343919.478	41214.468

MISSOZI SITE

ACCESS ROAD

RAILWAY LINE

320 ACRES



SITE LOCATION COORDINATES**PROJECTON:UTM 36N**

Datum: Code: Arc 1960 Description: Arc 1960, meanValue, Kenya, Tanzania Ellipsoid: CLRK80(Clarke 1880, RGS)

Points	Easting (M)	Northing (M)	Latitude (DD)	Longitude (DD)	TILT (degree)
P1	513133.52	314636.38	2.846584	33.118172	45
P2	512668.25	315187.87	2.851574	33.113986	45
P3	512523.15	315144.76	2.851184	33.112681	45
P4	512410.27	315145.00	2.851187	33.111665	45
P5	512386.90	315120.68	2.850967	33.111455	45
P6	512609.68	314654.72	2.846751	33.113459	45
P7	512960.69	314530.07	2.845623	33.116617	45
		AREA= 24.283 Ha 60.00 Acres			

LAYOUT DIAGRAM FOR LAND AT PADERKILAK PADER DISTRICT

